Original Contributions

Mammalian Genome
Incorporating Mouse Genome
From Genotype to Phenotype

Chromosome substitution strains: some quantitative considerations for genome scans and fine mapping

John K. Belknap

Research Service (R&D-5), Veterans Affairs Medical Center, and Department of Behavioral Neuroscience, Oregon Health & Science University, Portland, OR 97239, USA

Received: 6 January 2003 / Accepted: 8 July 2003

Abstract

A chromosome substitution strain (CSS) is an inbred strain in which one chromosome has been substituted from a different inbred strain by repeated backcrossing. A complete CSS set has one strain representing each chromosome against a uniform background, thus allowing genome-wide scans to be carried out for quantitative trait loci (QTLs) influencing any trait of interest. A one-way ANOVA by strain is first carried out, followed by planned comparisons using Dunnett's method. A QTL is detected and mapped to a chromosome when a significant difference is observed in a background strain vs CSS comparison. The most efficient ratio of background to CSS mice in any one comparison is 4.5:1, and the threshold for $p < .05$ genome-wide significance is estimated to be $p = .003$ to .004, a much less stringent criterion than any other mammalian mapping population. The use of false discovery rates tends to further reduce threshold stringency. Comparisons are made to the widely used conventional F_2 intercross, and both advantages and disadvantages are noted. The proportion of the trait variance due to a QTL is often much larger than the same QTL in an F_2, and the number of generations to attain fine mapping is greatly reduced. To serve as guidelines for planning experiments, methods to estimate sample sizes for QTL detection are presented for the initial genome scan and for subsequent fine mapping.

The limitations and expense of segregating populations for the genetic analysis of complex traits have spurred the development of several sets of inbred mouse strains as alternative mapping populations. These sets offer the many advantages of any inbred strain as well as obviating the need for genotyping for the initial genome scan. Among these are recombinant inbred strains (Williams et al. 2001), recombinant congenic strains (Demant and Hart 1986; Fortin et al. 2001), and most recently, chromosome substitution strains (CSS) (Nadeau et al. 2000). The advantages of CSS compared to the others have been summarized by Nadeau et al. (2000).

The term congenic strain, as used in mouse and rat genetics (Bailey 1981), is essentially equivalent to a substitution strain as used in the genetics of plant and most invertebrate species (Kearsey and Pooni 1996; Mackay 2001). Thus, the term CSS, referring to an entire substituted chromosome, is more in keeping with terminology used in most species other than laboratory rodents. The first CSS set to be developed in mice partitioned the donor strain (A/J) genome into individual chromosomes, each present in one of the 21 CSS against a C57BL/6J background (the B.A set; Nadeau et al. 2000). In addition to the 19 autosomes, the X and Y chromosomes are each represented by a CSS. Developing a new CSS set requires an extraordinary amount of work, time, and expense, but once developed, they are an enduring and tremendously valuable resource to the genomics research community.

A genome scan in the simplest case is comprised of a one-way ANOVA by strain followed by the comparison of each CSS with the background strain on a trait of interest. A significant difference for any one comparison indicates the presence of one or more QTLs on the differential chromosome. In this paper, attention is turned to the quantitative genetic aspects of using these newly developed strains for genome scans and for subsequent fine mapping. It is hoped that this initial exploration will encourage the development of even more powerful approaches than the relatively simple ones outlined below.

In what follows, an approximate normal distribution and a common variance among CSS are assumed. The heritability of the trait, h^2, or the

Correspondence to: J.K. Belknap; E-mail: belknajo@ohsu.edu

proportion of the trait variance due to all QTLs in the aggregate, can be subdivided into the effects of individual QTLs, or v^2, the proportion of trait variance due to an individual QTL. This quantity will also be referred to below as the QTL effect size. An individual CSS strain compared with its background strain is referred to as a CvsB comparison. Because there are no heterozygotes to show dominance, only the additive effects of QTLs are assessed. The CSS bearing the donor Y Chromosome (Chr) will not be directly considered because it accounts for such a tiny portion of the genome. This results in 20 CvsB comparisons to cover the genome rather than 21. However, because of the paucity of genes on this chromosome, this CSS can be included in a genome scan without appreciably changing any of the conclusions to be made below except for the complications of sex linkage at the fine mapping stage.

In most of the discussion below, it is assumed that 20 CSS strains exist, all with the same progenitor strain as background and the other progenitor as donor, e.g., the B.A set. However, reciprocal CSS sets with both progenitors as background, involving a total of 40 strains (e.g., B.A and A.B), offer several additional opportunities for genetic analysis (for example, epistasis), as is briefly discussed below. These are in the planning stage.

Because the F_2 between two inbred strains is the most common choice for genome scans, it is instructive to compare F_2 with CvsB scans when both populations are derived from the same two inbred strains. For this purpose, frequent reference will be made to an example of a QTL accounting for 6% of the trait variance in a conventional F_2 with $h^2 = .40$, typical values in the QTL literature. However, equations will be given or cited below that allow the use of any values of interest.

Methods and results

Thresholds for statistical significance for a CSS genome scan. The simplest approach to scan the genome is to compare CSS with the background strain for a total of 20 CvsB comparisons. Following accepted practice, the significant threshold (α_G) is set at a genome-wide $p = .05$ ($\alpha_G = .05$), which implies that the risk of even one false positive QTL is no greater than 5% in any one genome scan (Lander and Kruglyak 1995). In order to accomplish this, the significant threshold set for each of the individual CvsB comparisons, or α_S, can be estimated by using a Bonferroni method, as follows (Rice 1989):

$$\text{Equation 1: } \alpha_S = \alpha_G/k$$

where k is the number of comparisons; $k = 20$ in our case. When the genome-wide α_G is set at .05, the required α_S to accomplish this is $p = .05/20 = .0025$. In other words, $p = .0025$ should be the significant threshold for single CvsB comparisons to obtain a genome-wide $p = .05$ for all 20 comparisons.

The Bonferroni approach to correct for multiple comparisons presumes that each CvsB comparison is independent of the others, which is not strictly true in our case because the same background strain group is used repeatedly for all 20 comparisons. Dunnett (1955) worked out the thresholds to be set in this special case, where multiple groups (CSS in our case) are all compared with a single control group (background strain). The Dunnett approach to setting α_S when applied to 20 strains is approximated by substituting $k = 13$ in Equation 1 rather than $k = 20$ when the total sample size for all 20 comparisons is >120 (Hochberg and Tamhane 1987; Table 3). Dunnett's method yields $\alpha_S = .004$ instead of the .0025 estimate based on independent tests. This criterion is for a 1:1 ratio of background to CSS mice in each CvsB comparison. As will be discussed below, a 4.5:1 ratio is more efficient (requires fewer mice) than 1:1; and when a 4.5:1 ratio is used, $\alpha_S = .003$ (Hochberg and Tamhane 1987; Table 3). Note that the $p = .003$ to .004 criterion for genome-wide significance is much less stringent than any other mouse mapping population (Lander and Kruglyak 1995). The primary reason for this is the lack of recombination owing to crossing-over in CvsB comparisons.

It is likely that most CSS scans will be followed by higher resolution mapping efforts for the more promising chromosomes, which in turn will require intercrossing or backcrossing of selected CSS with the background strain coupled with genotyping and phenotyping of the derived populations (see below). These follow-up crosses provide additional information concerning the existence of a QTL to supplement that gained from the initial genome scan. In this case, both the initial CSS scan and follow up crosses can be used jointly to test for significance (two- or multi-step), a much more powerful test than the genome scan alone. For example, Fisher's method can be used to combine probabilities across the two (or more) independent experiments testing the same hypothesis about the existence of a QTL (Sokal and Rohlf 1995). The rationale for pooling QTL results across independent experiments in tests for significance has been discussed by Belknap and Atkins (2001). This approach allows a relaxed criterion to be used for the genome scan alone, for example, the suggestive rather than the significant criterion. The suggestive criterion according to Lander and Kruglyak (1995) is that value of α_S that yields an average of one false positive in a genome scan. The α_S value for single CvsB comparisons that accomplish this is $p = .05$, because 20 comparisons at $p = .05$ will yield, on average, one false positive ($.05 \times 20$) over all 20 comparisons when the null hypothesis (no QTLs exist) is true. For example, using Fisher's method, $p = .05$ for the initial scan can be combined with $p < .01$ obtained from one or more of the follow-up crosses to obtain a combined $p < .004$, a significant outcome. Relaxing the α_S criterion for the initial scan in this way has another important advantage, an increase in power to detect QTLs for a given sample size.

The classical method of correcting for multiple testing is to set α_S for single tests so as to obtain a genome-wide $p < .05$ for all tests (Miller 1981). The Bonferroni and Dunnett's methods presented above and the guidelines offered by Lander and Kruglyak (1995) are all examples of this strategy. A different and promising method is to correct for multiple testing by using false discovery rates (FDR). The false discovery approach is based on estimating the proportion of all significant tests that are expected to be false positives rather than on what proportion of all tests are expected to be false positives (Benjamini and Hochberg 1995; Storey 2002; see http://faculty.washington.edu/~jstorey/qvalue/

While the precise calculation of FDR is rather complex (Storey 2002), a simple approximation can be gained by the following general expression: FDR $\approx \alpha_S (N_{\text{tests}}/N_{\text{significant tests}})$, where α_S is the threshold p value used to judge significance of each of N_{tests}, and $N_{\text{significant tests}}$ is the number of tests observed to exceed this α_S threshold (Lee et al. 2002). Rearranging this expression gives us the α_S value that will result in any desired FDR as follows: $\alpha_S \approx (\text{FDR} \times N_{\text{significant tests}})/N_{\text{tests}}$. For example, if we observe four significant tests out of 20 CvsB comparisons, a 5% FDR is obtained by setting $\alpha_S \approx (.05 \times 4/20) = .01$, a less stringent significant criterion than the .003 to .004 obtained using Dunnett's method. If the FDR is set in advance (usually at .05 or .01), α_S cannot because it must be adjusted according to the number of significant tests actually observed in an experiment. Comparing the 5% genome-wide criterion with the 5% FDR, they are equivalent when there is only one significant test, but the false discovery method results in less stringent significant thresholds when there are two or more significant tests. This results in a gain in power to detect QTLs, but at the expense of a higher rate of false positives.

Comparison of the effect size of a given QTL expected in an F_2 vs CvsB. The proportion of the trait variance due to a given QTL in a conventional F_2 is symbolized as $v^2_{F_2}$, and in a CSS vs Background strain comparison (CvsB), as v^2_{CvsB}. When the same QTL in the two populations is compared, the following shows their expected relationship when there is no dominance and the numbers of CSS and background strain mice in a CvsB comparison are approximately equal:

$$\text{Equation 2: } v^2_{\text{CvsB}} = 2v^2_{F_2}/(1 - H^2)$$

The factor of 2 in the numerator reflects the doubling of the QTL effect size in the CvsB because only homozygotes are present, whereas in an F_2, half the population are heterozygotes that contribute nothing to the QTL effect size in the absence of dominance. H^2 is the residual heritability for the trait in the F_2 (Darvasi 1998), or the heritability of all QTLs other than the one we are considering. H^2 is zero in the CvsB comparison because the rest of the genome is the same in both strains, but is nonzero in a conventional F_2 owing to the segregation of additional QTLs elsewhere in the genome.

While Equation 1 above presumes no dominance, complete dominance can be accounted for by changing the factor of 2 in the numerator to 1.33 (Bruell 1971). Equations for any degree of dominance are given by Bruell (1971) and Belknap and Atkins (2001).

Note that v^2_{CvsB} represents the combined effect of all QTLs on a given chromosome. Nonetheless, for simplicity, it will be provisionally assumed that only one QTL is present in much of the discussion to follow. Of course, any number of loci could underly a QTL detected in a CvsB comparison, but if they present as a single statistical entity, the singular is used below.

With our example, it is assumed that a trait has a heritability in an F_2 of $h^2 = .40$ and a particular QTL has $v^2_{F_2} = .06$. H^2 is then equal to .40 minus .06 = .34, and v^2_{CvsB} from Equation 2 is estimated to be .18. This value is three times larger than the F_2 value for the same QTL. If there is complete dominance, the estimated v^2_{CvsB} is .12, double the F_2 value. In general, the effect of a given QTL in a CvsB comparison will be larger than in an F_2, and the magnitude of the difference depends on whether dominance is present and on the magnitude of H^2. This effect size advantage greatly aids QTL detection by the CvsB method.

While Equation 2 presumes a 1:1 ratio of background to CSS mice in a CvsB comparison, other ratios can be accommodated by multiplying the estimate of v^2_{CvsB} from Equation 2 by $4PQ$, where P and Q are the proportion of mice in each CvsB comparison from the background strain and CSS, respectively.

Statistical analysis of CSS data. The first step is to carry out a one-way analysis of variance (ANOVA) by CSS for a trait of interest across all 20 CSS and the background strain. A significant F value implies that there is significant additive genetic variation in the population. Moreover, R^2, the sum of squares between strains divided by the total sum of squares, gives an estimate of the heritability of the trait (h^2), or the proportion of the trait variance due to additive genetic influences (narrow sense heritability). Also of value from an ANOVA table is the mean square within strains (MSw), which is a pooled estimate across all strains of the within-strain (environmental) variance, also known as the error variance. MSw is then used in carrying out specific CSS vs background strain (CvsB) comparisons by using a standard two-sample t test shown in Equation 3, where N_{CSS} and N_{B} are the numbers of background and CSS mice in a CvsB comparison, respectively, and $N_{\mathrm{CvsB}} = N_{\mathrm{CSS}} + N_{\mathrm{B}}$.

$$\text{Equation 3: } t = \frac{\mathrm{Mean_{CSS}} - \mathrm{Mean_{Bkgrd}}}{\sqrt{[(N_{\mathrm{CvsB}}/N_{\mathrm{CSS}}N_{\mathrm{B}})MS\mathrm{w}]}}$$

The degrees of freedom (df) for the t test is also the df for the MSw from the one-way ANOVA; thus, it may be symbolized as dfw. Because df will exceed 100 in almost all cases involving complex traits (testing six mice per strain or more will yield $df > 100$), we can use the normal variate z rather than t for simplicity. The threshold value of t (or z) to attain significant ($p = .004$) or suggestive ($p = .05$) status is, therefore, 2.9 and 1.96, respectively. For the FDR method, these values would depend on the number of observed significant tests.

Estimating v^2_{CvsB} directly from CvsB comparison data: While Equation 2 allows the estimation of v^2_{CvsB} from conventional F_2 data, of more practical importance is an estimate from the results of a CvsB comparison. The simplest way to accomplish this is to use Equation 4 (Rosenthal 1994; Eq. 16–18):

$$\text{Equation 4: } v^2_{\mathrm{CvsB}} = t^2/(t^2 + N_{\mathrm{CvsB}} - 2)$$

where t and N_{CvsB} are from Equation 3.

Estimated sample size to detect a QTL of a given effect size. Equation 5 below yields an estimate of the sample size of the CSS (N_{CSS}) and background (N_{B}) to detect a QTL of effect size v^2_{CvsB} in a CvsB comparison (Belknap 1998) comprised of proportion P background strain mice and Q CSS mice.

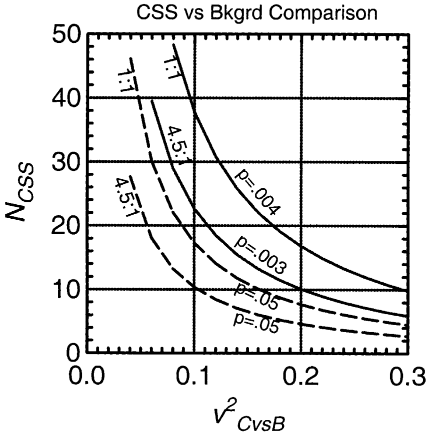

Fig. 1. Estimates of the CSS sample size (N_{CSS}) to detect a QTL with effect size v^2_{CvsB} in a CvsB comparison are plotted for the significant ($p = .003$ or .004, solid lines) and suggestive ($p = .05$, dotted lines) thresholds for a genome scan using Equation 5. It is assumed that the ratio of background strain to CSS mice in a CvsB comparison is either 1:1 or 4.5:1, as shown. If the ratio is 4.5:1, the N_{CSS} values shown should be multiplied by 4.5 to get N_{B} (background strain) sample size estimates. It is presumed that the statistical power is set at 0.5. If power = 0.8 is desired, the N_{CSS} values (and from those the N_{B} values) given by the $p = .003$ or .004 curves should be multiplied by 1.7, and the $p = .05$ curves by 2.25. For the FDR method, see text.

$$\text{Equation 5: } N_{\mathrm{CSS}} = \frac{1}{2}Z\alpha^2(1 - v^2_{\mathrm{CvsB}})(4PQ)/(v^2_{\mathrm{CvsB}})$$
$$N_{\mathrm{B}} = N_{\mathrm{CSS}}(P/Q)$$

$Z\alpha$ is the normal variate for the desired α_{S} value, either 2.9 for significant or 1.96 for suggestive, and v^2_{CvsB} is the proportion of the trait variance due to a QTL in a CvsB comparison as defined above. Figure 1 shows the relationship among these variables when a 1:1 ratio is used, i.e., $N_{\mathrm{B}} = N_{\mathrm{CSS}}$, thus $P = Q = .5$. In our reference example where $v^2_{\mathrm{CvsB}} = .18$, $N_{\mathrm{CSS}} = 20$ mice (rounding up) are needed to obtain significance ($Z\alpha = 2.9$). Thus, the estimated total number of mice needed in a CvsB comparison ($N_{\mathrm{B}} + N_{\mathrm{CSS}}$) would be twice this or 40 mice to detect this particular QTL effect size at the significant threshold. For the FDR method, the value of $Z\alpha$ would vary as a function of the number of observed significant tests, and thus so would the sample sizes needed to obtain a significant result.

Equation 5 presumes that the statistical power to detect a QTL is 0.5, which provides a reasonable but somewhat minimal estimate. Any other desired power can be estimated by substituting $(Z\alpha + Z\beta)^2$ in place of $Z\alpha^2$ in Equation 5, where $Z\beta$ is the normal variate value (one-tailed) corresponding to the Type II (false negative) error risk one is willing to assume. For example, when power = 0.8, the Type II error risk is thus 1 − power = 0.2, and $Z\beta$ is, then, 0.85 when df >100. If Power = 0.90 is desired, $Z\beta$ = 1.28 should be used.

When carrying out a genome scan, however, testing larger numbers of background strain mice than CSS mice in a CvsB comparison is more efficient (requires fewer mice). Another advantage is to more readily allow concurrent testing of background and CSS mice throughout a typical several-week testing period to correct for any environmental changes. Any higher ratio can be used in Equation 5 by inserting the appropriate values of P and Q, the proportions of mice used in a CvsB comparison that come from the background and CSS, respectively.

What ratio is most efficient of animal numbers? When there are multiple groups compared with a single control group, Bechhofer and Tamhane (1983) offer a complex solution to this question, but it is closely approximated when df > 100 by a square root allocation rule, as follows: $N_B = N_{CSS}\sqrt{k}$ (Dunnett 1955), where k = number of comparisons (i.e., 20). This suggests a ratio of $\sqrt{20}$ to 1, or 4.5 to 1, as the most efficient ratio of background to CSS mice in any one CvsB comparison used as part of a genome scan. Figure 1 shows the estimated numbers of CSS mice (N_{CSS}) needed when a 4.5:1 ratio is used based on Equation 5 for both the significant and suggestive thresholds. For comparison, the values for a 1:1 ratio are also plotted.

Using the square root allocation rule, we can estimate the optimal value of P using the equation $P = \sqrt{k}/(\sqrt{k}+1)$, which for a 4.5:1 ratio gives us P = 4.5/(4.5 + 1) = .82, $Q = 1 - P$ = .18. Placing these values into Equation 5 gives us N_{CSS} = 12 mice, which is smaller than the 20 mice needed when a 1:1 ratio is used. The background strain numbers (N_B) would then be P/Q or 4.5 times greater than N_{CSS}, or 4.5 × 12 = 54 mice. This one group of background strain mice is then used in all 20 CvsB comparisons. By adopting a 4.5:1 ratio calculated in this way, the numbers of CSS mice used in any CvsB comparison can be reduced to 60% of the mice needed with a 1:1 ratio (from 20 to 12), yet $Z\alpha$ or any particular value of z (or t) would remain the same as expected with a 1:1 ratio when df >100 for the t test given in Equation 3. Put another way, the power of the test to detect QTLs remains approximately the same across a range of ratios when calculated as described above. However, to accomplish this, the total number of mice in each CvsB comparison must be increased from 40 (1:1) to 66 (4.5:1). This is inefficient (requires more mice) if only one CvsB comparison was made, but summing across all 20 CvsB comparisons needed for a genome scan, it is more efficient. In our example, the total number of mice decreased from 420 mice for 1:1 (20 CSS strains × 20 mice + 20 background strain mice) to 294 mice for 4.5:1 (20 CSS strains × 12 mice + 54 background strain mice). In other words, the 294 mice at 4.5:1 give the same power to detect QTLs genome-wide as does 420 mice at 1:1. The 4.5:1 ratio is not only more efficient than 1:1, it is also the most efficient of any ratio based on 20 comparisons (Bechhofer and Tamhane 1983). However, a range of ratios from 3:1 to 7:1 yield total sample sizes within 3% of the most efficient ratio, so the precise ratio selected within this range is not critical.

From the foregoing, we can also estimate the threshold difference between the CSS and background strain means expressed in within-strain SD units (δ) needed to attain significance for any ratio, as follows: $\delta = t\sqrt{(N_{CvsB}/N_B N_{CSS})} = 2.9 \times .316 = .92$ SD units (Rosenthal 1994; Eq. 16–21), where t, N_{CvsB}, N_B, and N_{CSS} are the values at the significant threshold for our example given above. For the suggestive threshold, δ = 1.96 × .316 = .62 SD units. For the FDR method, the t value that yields a 5% FDR would be used, which will vary depending on the number of significant tests observed.

Earlier it was concluded that the threshold for statistical significance using Dunnett's method was p = .004 obtained from $Z\alpha$ = 2.9 (two-tailed). This is true for the 1:1 ratio, but with a 4.5:1 ratio, $Z\alpha$ becomes about 3.0 (Hochberg and Tamhane 1987; Table 3) and the significant threshold becomes p = .003. A similar small adjustment would be needed if the FDR method is used.

Finally, it should be noted that there are other options for CSS QTL analysis other than the relatively simple ones presented here. Because somewhat greater power can be gained, they should be explored for their utility. For example, there are a number of different stepwise procedures where the outcome of one CvsB comparison can affect the outcome of subsequent ones (e.g., Rice 1989; Hochberg and Tamhane 1987). Stepwise linear contrast models may also hold promise where comparisons among CSS are also used as well as with the background strain. Because of their much greater complexity, and in some cases, the greater need for caution, these approaches are not reviewed here.

Higher resolution mapping. Once a CSS has been identified that harbors a QTL, a cross to the background strain is carried out to produce an F_1, followed by a congenic backcross (ConBC), or intercross (ConF2). [The adjective congenic is used here because recombination will reduce the size of the introgressed region to less than the entire chromosome when fine mapping is the goal.] Both the ConBC and ConF2 are segregating only for the loci on a portion of one chromosome, which facilitates higher resolution mapping because the effects of other QTLs segregating at other chromosomes (H^2) have been eliminated.

There are two main approaches to attaining higher resolution mapping from these populations, which after Darvasi (1998) are the interval specific congenic strain (ISCS) and the recombinant progeny testing (RPT) approaches. These two were the most efficient of all fine mapping (to 1 cM) methods reviewed by Darvasi. The first approach, also known as congenic mapping or genetic chromosome dissection, is well exemplified by the work of Cicila et al. (2001) and Denny et al. (1997) among others, and the second approach by Fehr et al. (2002) and Bennett et al. (2002). While these studies used congenic (substitution) strains rather than a CSS as their point of departure, the methods and overall strategy remain much the same.

Both approaches are similar in that they rely on one or more segregating generations to generate a high density of recombinations, and both rely on replicating each recombinant mouse found in one generation so that there are several to many mice with the same recombination in the next generation for phenotyping and genotyping. The pattern of which recombinant genotypes show the QTL effect, and which do not, allows higher resolution mapping (see Fig. 2). Central to this effort is the capacity to detect a QTL effect, which in turn depends on QTL effect size and on sample size. We turn our attention to these two considerations below.

The two approaches differ in that with ISCS, the goal at the outset is to derive a new set of congenic strains from a ConF2 whose introgressed regions differ, while with RPT higher resolution mapping is done in a series of segregating backcrosses or intercrosses, and a final subcongenic strain is produced only after fine mapping has been achieved. Another difference is that, with ISCS, the background strain chromosome is represented by the background strain itself, but with RPT, it is represented by ConBC or ConF2 mice possessing the background strain chromosome (or portion thereof) who are littermates of mice possessing the donor strain segment. Because with RPT the genotypes compared involve littermates from the same mothers, genetically mediated maternal effects are largely eliminated as a source of phenotypic differences. In contrast, the ISCS vs background strain comparison (and also the CvsB comparison) involves different mothers of different genotypes, thus allowing maternal effects (if any) to become part of the phenotype under study.

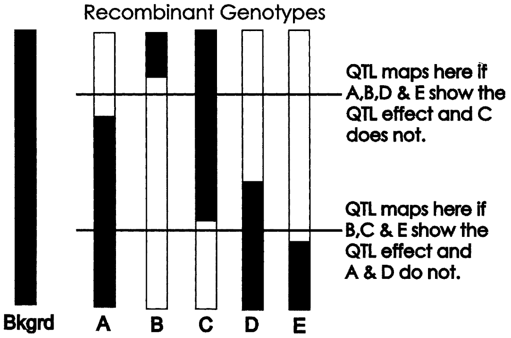

Fig. 2. Higher resolution mapping begins by generating unique recombinations in a substituted chromosome, each occurring initially in a single mouse, followed by replication in the next generation(s) to produce many mice with the same recombination. The white and black areas in the figure denote regions of one chromosome originating from the donor and background strains, respectively. The pattern of which replicated offspring show the QTL effect and which do not (when compared with the background strain) yields higher resolution mapping. To illustrate this, five replicated genotypes are shown, labeled A through E, which can be either interval-specific congenic strains (ISCS), where all regions shown are homozygous, or recombinant progeny testing (RPT) offspring from a backcross (ConBC), where the white areas denote donor strain heterozygous regions compared with black areas denoting background strain homozygous regions. A congenic F_2 (ConF2) is more complex and is thus not shown for clarity. This approach is then extended into subsequent generations of either ISCS or RPT (not shown)–additional recombinants are sought and replicated in an ever-narrowing QTL confidence interval until the desired map resolution is attained.

ISCS approach. Once the new set of congenic strains (ISCS) is formed from a ConF2 with each strain representing a unique recombination of the substituted chromosome, the strains are phenotyped, and the pattern of which ISCS show the QTL effect (differ from the background strain), and which do not, yields higher resolution mapping (Darvasi 1997; see Fig. 2). Since the detection of the QTL effect involves a congenic vs background strain comparison, almost all of what was presented earlier about CSS vs background strain (CvsB) comparisons applies here as well. Concerning statistical significance, a one-tailed α_S = .05 should suffice whenever the existence and direction (which allele predisposes to higher trait scores) of the QTL have been previously established at the significant criterion. In this case, we can use the p = .05 plot in Fig. 1 for a 1:1

ratio (dotted line), which is two-tailed, and estimating the corresponding one-tailed values by decreasing the N values shown in that figure by 29%.

Finally, because several ISCS are usually compared to the background strain, sample size ratios higher than 1:1 (background strain: congenic strain) can be used to advantage as was the case for CvsB comparisons discussed previously. The square root allocation rule can also be used here to select a ratio (Dunnett 1955).

RPT approach. Each recombinant mouse found in any one segregating generation is backcrossed or intercrossed so that several to many mice are produced with the same recombination in the next generation. These recombinant progeny are genotyped and phenotyped, and higher resolution mapping is attained from the pattern of which recombinant progeny show the QTL effect, and which do not (Fig. 2).

With RPT, it is advantageous to start with a congenic F_2 (ConF2) to determine the magnitude and direction of dominance, which is important in deciding which crosses to use in the subsequent generations needed for fine mapping. The QTL effect size (v^2) in a ConF2 or ConBC, when compared with the same QTL seen in a CvsB comparison (v^2_{CvsB}), is expected to be as follows when the cofactors f_1 and f_2 are as listed below:

Equation 6:

$$v^2 = \frac{f_1 v^2_{\text{CvsB}}}{(1 - f_2 v^2_{\text{CvsB}})(4PQ)}$$

	f_1	f_2
ConF2, no dominance	$\frac{1}{2}$	$\frac{1}{2}$
ConF2, complete dom.	$\frac{3}{4}$	$\frac{1}{4}$
ConBC, no dominance	$\frac{1}{4}$	$\frac{3}{4}$
ConBC, complete dom.	1	0

For example, if $v^2_{\text{CvsB}} = .18$, as given in our recurring example from Equation 2, then $v^2 = .10$ is expected in an ConF2 with no dominance when $P = Q = .5$. This compares to an effect size of .06 for the same QTL seen in a conventional (not congenic) F_2 noted above. This difference is due to the inbred vs segregating nature of the rest of the genome. Once we have our estimate of v^2 for either a ConF2 or ConBC derived from CvsB comparison data, the sample size (N_C) to detect a QTL can be estimated to be:

$$\text{Equation 7: } N_C = Z\alpha^2(1 - v^2)/v^2$$

When the existence of a QTL has been established previously, $\alpha = .05$ (one-tailed) is acceptable for QTL detection, thus $Z\alpha = 1.65$. Then, based on Equation 7, the number of ConF2 mice needed to detect the QTL when $v^2 = .10$, for example, is estimated to be $N_C = 24$ mice when there is no dominance and $N_C = 16$ mice when there is complete dominance. This presumes power = 0.5, but any other power can be accommodated by using the correction described above for Equation 5.

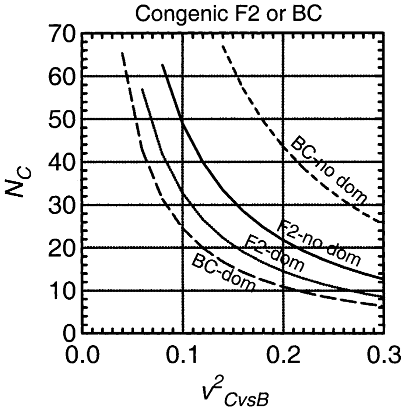

Fig. 3. The estimated sample size (N_C) needed to detect a QTL in a congenic F_2 or congenic backcross is plotted as a function of v^2_{CvsB} observed in a CvsB comparison based on Equations 6 and 7 combined. It is assumed that the detection threshold is $p = .05$ (one-tailed) for a QTL whose existence and direction of effect has been established in a prior genome scan. For a ConBC, it is assumed that the donor strain allele is completely dominant over the background strain allele, or there is no dominance. When the direction of dominance is reversed, i.e., the background strain allele is completely dominant, a ConBC is not effective, and thus is not shown.

From an observed value of v^2_{CvsB} from a CvsB comparison, estimates of N_C for either a congenic F_2 or BC are shown in Fig. 3 based on Equations 6 and 7 combined. Plots for the presence and absence of dominance are shown. For a ConBC, it is assumed that the direction of dominance is always in the advantageous direction, i.e., the donor strain allele is dominant (the background strain allele is recessive) or there is no dominance. When partial dominance exists in a ConBC (the dominance deviation equals one-half of the additive effect of an allele substitution), the ConF2 no dominance plot in Fig. 3 is appropriate.

In a ConBC, when dominance is in the disadvantageous direction, that is, the background strain allele is dominant (donor allele is recessive), the QTL cannot be detected because the two genotypes at a QTL will not differ phenotypically and thus cannot contribute to the trait variance (Bruell 1971; Belknap and Atkins 2001). In this case, backcrosses to the other progenitor strain should be used if the CSS set offers both progenitors as background. If the CSS set is based on only one progenitor as background, and the direction of dominance is dis-

Table 1. Comparison of a genome scan with either a conventional F_2 or 20 CvsB (CSS vs background strain) comparisons derived from the same progenitors. It is assumed that the QTL to be detected accounts for 6% of the F_2 trait variance, with no dominance, and the heritability of the trait is .40

	F_2	CvsB
• Heritability of QTL (v^2):	.06	.18[a]
• Significant threshold:	p = .0001	p = .003[b]
• Suggestive threshold:	p = .0025	p = .05
• N to detect significant QTLs:	N = 327[c] (1/4 genotyped)	N = 294[d] [54 bkgrd strain; 12 CSS per CvsB]
• N to attain suggestive QTL:	N = 210[c] (1/4 genotyped)	N = 150[d] [27 bkgrd strain; 6 CSS per CvsB]
• Genotypings for genome scan: (80 markers × ¼ × 327 mice)	5,800	None
• Generations to obtain a substitution (congenic) strain:	6[e]	0
• Generations to obtain 1–2 cM map resolution (ISCS or RPT):	9–10[e,f]	3–4[f]
• 95% confidence interval for a QTL at the significant threshold:	50 cM[g]	Syntenic

[a] Assumes a 1:1 ratio of background to CSS mice in each CvsB comparison. With a 4.5:1 ratio, v^2 = .11 rather than .18. If complete dominance exists, v^2 = .12 rather than .18.
[b] Assumes Dunnett's method of correcting for multiple tests. For the FDR method, see text.
[c] Total N for a genome scan estimated from Equation 7 followed by the Darvasi and Soller (1997) correction for selective genotyping. With full genotyping, N =237 conventional F_2 mice are needed for the significant threshold and N =150 for suggestive, but the number of genotypings would then be fourfold greater than shown in the table.
[d] Assumes a 4.5:1 ratio of background to CSS mouse numbers (see text).
[e] Assumes a ''speed'' congenic method is used involving five backcross generations followed by an intercross to produce a congenic (substitution) strain (Wakeland et al. 1997).
[f] Assumes that three to four generations are needed to attain 1–2 cM map resolution by ISCS or RPT starting with a congenic (substitution) strain (e.g., Fehr et al. 2002). While this could be done in one generation, it is more efficient to use a sequential approach where the QTL confidence interval becomes progressively smaller over two or more generations.
[g] Estimated by using the method of Darvasi and Soller (1997).

advantageous, then an ConF2 intercross strategy would have to be used for fine mapping. This is because, in contrast to a ConBC, the direction of dominance in a ConF2 does not affect the QTL effect size or N_C estimates.

Discussion

Among populations used for genome scans, there is a trade-off between map precision and the stringency of the α_S level required to correct for multiple testing. At one extreme is a simple two inbred strain comparison that requires no correction for multiple testing; thus, relatively small sample sizes can detect significant differences, but it can only detect the effects of all QTLs in the aggregate, with no information at all about location. On the other extreme are very advanced intercross lines, such as the HS stock, that have an extremely high density of recombinations accumulated over many generations. These provide high map precision (Talbot et al. 1999), but would require a very stringent α_S level ($p \sim 1 \times 10^{-6}$) if used for a genome scan. The conventional F_2 is intermediate, while CSS fall between the F_2 and two inbred strain comparisons on this trade-off continuum.

A CSS set offers a very simple structure of the genome—one strain, one chromosome. Because of this, relatively simple statistical tools can be quite effective and efficient for detecting QTLs. An attempt has been made throughout this article to maintain this relative simplicity by avoiding complex analyses when simple ones will do. This is in contrast to most other QTL mapping populations (F_2, BC, RI, RCS) which often require the use of specialized software implementing complex algorithms beyond the understanding of many users. In contrast, even a rudimentary statistics computer package is sufficient for a genome scan with the methods presented here, and the power to detect QTLs is close to that attainable using much more complex analyses such as those briefly described in Methods. However, more complex analyses should be explored when they offer advantages, a subject of future papers.

Using a CSS set offers other advantages for both the initial genome scan and for subsequent higher resolution mapping when compared with a conventional F_2. Most of these have been described by Nadeau et al. (2000) and thus won't be repeated here. From the present analysis (Table 1), CSS are much more economical because genotyping costs are much reduced and because fewer generations are required to attain the higher map resolution often required for gene identification. Maintaining 21 strains is within the capacity of most laboratories, thus allowing multiple use across multiple traits, each in an independent sample if necessary. The stability and specifiability of these genotypes also allow the accumulation of information across laboratories and time, much as is the case for any set of inbred strains. Segregating populations, in contrast, are often single-use populations that are discarded when a genome scan is complete, and are useful only for the number of traits that can be reasonably tested on each mouse without some tests confounding the results of those to follow (carryover effects).

There are several disadvantages in the use of CSS. In the initial genome scan, map resolution is low (syntenic only) and only additive effects of QTLs are screened, thus dominance (intralocus interactions), and epistasis (interlocus interactions) do not contribute to the capacity for QTL detection. Higher resolution mapping and dominance variation do enter the picture in the follow-up crosses, and if both progenitors were used as background in a CSS set, background effects (a form of epistasis) can be efficiently studied where the effect size of a QTL depends on the genetic background. While epistasis other than background effects is not a factor in the initial scan, crosses between two CSS, each harboring one of two interacting loci, is a very powerful tool to assess epistasis and to fine map the interacting loci against an inbred background. To know which CSS to use in this way often presupposes that the interaction was detected with another mapping population from the same progenitors such as a conventional F_2, RI set, or recombinant congenic set. This is one reason why the use of more than one mapping population derived from the same progenitors is often advantageous when each provides information the others lack or provide to a lesser degree, an issue discussed by Bergeson et al. (2001). For this reason, combinations of different mapping populations can be more powerful than any one population alone. However, if one is limited to only one mapping population, at least initially, CSS have very much to offer in all phases of complex trait analysis from QTL detection to fine mapping to gene identification.

Acknowledgments

This work was supported by grants AA10760, AA06243, DA10913, DA05228, Merit Review Program #350, and a Senior Career Scientist Award from the Department of Veterans Affairs.

References

1. Bailey DW (1981) Recombinant inbred strains and bilineal congenic trains. In: Foster HL, Small JD, Fox JG (eds), *The mouse in Biomedical Research*. Vol I., (New York: Academic Press), pp 223–239
2. Bechhofer RE, Tamhane AC (1983) Design of experiments for comparing treatments with a control: tables of optimal allocation of observations. Technometrics 25, 87–95
3. Belknap JK (1998) Effect of within-strain sample size on QTL detection and mapping using recombinant inbred mouse strains. Behav Genet 28, 29–37
4. Belknap JK, Atkins AL (2001) The replicability of quantitative trait loci (QTLs) for murine alcohol preference drinking behavior across eight different studies. Mamm Genome 12, 893–899
5. Benjamini Y, Hochberg Y (1995) Controlling the false discovery rate: a practical and powerful approach to multiple testing. J. R. Stat Soc B 57, 289–300
6. Bennett B, Beeson M, Gordon L, Carosone-Link P, Johnson TE (2002) Genetic dissection of quantitative trait loci specifying sedative/hypnotic sensitivity to ethanol: mapping with interval-specific congenic recombinant lines. Alcohol Clin Exp Res 26, 1615–1624
7. Bergeson SE, Helms ML, O'Toole LA, Jarvis MW, Hain HS et al. (2001) Quantitative trait loci influencing morphine antinociception in four mapping populations. Mamm Genome 12, 546–553
8. Bruell JH (1971) Dominance and segregation in the inheritance of quantitative behavior in mice. In: Manosevitz M, Lindzey G, Thiessen DD (eds) *Behavioral Genetics: Method and Research* (New york: Appleton-Century-Crofts), pp 92–109
9. Cicila GT, Garrett MR, Lee SJ, Liu J, Dene H et al. (2001) High-resolution mapping of the blood pressure QTL on Chromosome 7 using Dahl rat congenic strains. Genomics 72, 51–60
10. Darvasi A (1997) Interval-specific congenic strains (ISCS): an experimental design for mapping a QTL into a 1-centimorgan interval. Mamm Genome 8, 163–167
11. Darvasi A (1998) Experimental strategies for the genetic dissection of complex traits in animal models. Nat Genet 18, 19–24
12. Darvasi A, Soller M (1992) Selective genotyping for determination of linkage between a marker locus and a quantitative trait locus. Theor Appl Genet 85, 353–359
13. Darvasi A, Soller M (1997) A simple method to calculate resolving power and confidence interval of QTL map location. Behav Genet 27, 125–132
14. Demant P, Hart AM (1986) Recombinant congenic strains—a new tool for analyzing genetic traits determined by more than one gene. Immunogenetics 24, 416–422
15. Denny P, Lord CJ, Hill NJ, Goy JV, Levy ER et al. (1997) Mapping of the IDDM locus *idd3* to a 0.35 cM interval containing the *Il-2* gene. Diabetes 46, 695–700
16. Dunnett CW (1955) A multiple comparison procedure for comparing several treatments with a control. J Am Stat Assoc 50, 1096–1121

17. Fehr C, Shirley RL, Belknap JK, Crabbe JC, Buck KJ (2002) Congenic mapping of alcohol and pentobarbital withdrawal liability loci to < 1 centiMorgan interval of murine chromosome 4: identification of *Mpdz* as a candidate gene. J. Neuros 22, 3730–3738
18. Fortin A, Diez E, Rochefort D, Laroche L, Malo D, Rouleau GA, Gros P, Skamene E (2001) Recombinant congenic strains derived from A/J and C57BL/6J: a tool for genetic dissection of complex traits. Genomics 74, 21–35
19. Hochberg Y, Tamhane AC (1987) *Multiple Comparison Procedures*. (New york: J. Wiley & Sons)
20. Kearsey MJ, Pooni HS (1996) *The Genetical Analysis of Quantitative Traits*, (London: Chapman and Hall)
21. Lander ES, Kruglyak L (1995) Genetic dissection of complex traits: Guidelines for interpreting and reporting linkage results. Nat Genet 11, 241–247
22. Lee H, Dekkers JCM, Soller M, Malek M, Fernando RL, Rothschild MF (2002) Application of the false discovery rate to quantitative trait loci interval mapping with multiple traits. Genetics 161, 905–914
23. Mackay TFC (2001) The genetic architecture of quantitative traits. Ann a Rev Genet 225, 303–339
24. Miller RG (1981) *Simultaneous Statistical Inference*. (New york: Springer Verlag)
25. Nadeau JH, Singer JB, Matin A, Lander ES (2000) Analyzing complex genetic traits with chromosome substitution strains. Nat Genet 24, 221–225
26. Rice WR (1989) Analyzing tables of statistical tests. Evolution 43, 223–225
27. Rosenthal R (1994) Parametric measures of effect size. In: Cooper H, Hedges LV (eds) *The Handbook of Research Synthesis*, (New york: Russell Sage), pp 231–244
28. Sokal RR, Rohlf FJ (1995) *Biometry*, (San Francisco, CA: Freeman), pp 794–797
29. Storey JD (2002) A direct approach to false discovery rates. J R Stat Soc B 64, 479–498
30. Talbot CJ, Nicod A, Cherny SS, Fulker DW, Collins AC, Flint J (1999) High resolution mapping of quantitative trait loci in outbred mice. Nat Genet 21, 305–308
31. Wakeland E, Morel L, Achey K, Yui M, Longmate J (1997) Speed congenics: a classic technique in the fast lane (relatively speaking). Immunol Today 18, 472–477
32. Williams RW, Gu J, Qi S, Lu L (2001) The genetic structure of recombinant inbred mice: high-resolution consensus maps for complex trait analysis. Genome Biol 2:research0046.1.